

Thesis Topic Registration Form

Student's Data:

Student's Name: Fodor Tamás

Student's Neptun code: UVPXAG

Educational Information:

Training programme: programtervező informatikus, mesterképzés (MA/MSc)

I have an internal supervisor

Internal Supervisor's Name: Szalontai Balázs

Supervisor's Home Institution: Department of Software Technology and Methodology

Address of Supervisor's Home Institution: 1117, Budapest, Pázmány Péter sétány 1/C.

Supervisor's Position and Degree: tanársegéd, MSc

Thesis Title: Source Code Translation Between Programming Languages Using Deep Learning

Topic of the Thesis:

(Upon consulting with your supervisor, give a 150-300-word-long synopsis of your planned thesis.)

The aim of this thesis is to explore the use of deep learning models for source code translation between different programming languages. This research focuses on developing a model capable of converting source code written in one programming language into a functionally equivalent form in another language. The deep learning techniques considered for this task may include encoder-decoder transformers, recurrent neural networks and large language models. The research involves evaluating these different approaches to determine the most effective model for code translation.

To achieve this translation task, a dataset containing functionally equivalent pairs of programs will be used for training and evaluating the model. To assess code quality and evaluate the model's performance, various metrics will be applied, such as the CodeBLEU metric and functional tests to verify the quality and correctness of the translated code.

The goal of the research project is to achieve precise and efficient code transformation between different programming languages, aiming for the translated code to function correctly in the target language. Additionally, a basic software tool will be developed to illustrate the practical application of the deep learning model, allowing users to see the result of the translation.

Budapest, 2024. 11. 05.